DEEP-SPACE FLIGHT DIRECTIVE

EDGE OLYMPICS 2026

ORBITAL DEEP LEARNING OPTIMIZATION CHALLENGE
TFLITE HARDWARE PROTOCOL

OPERATION BLIND-NAV

LAUNCH WINDOW
30 Oct 2026
09:00 – 16:00

MISSION CONTROL
B316, NIT Warangal
Crew of 1 – 4

FLIGHT SILICON
Raspberry Pi 4B
2GB · ARM64

BOUNTY & COMMS
INR 5K / 3K / 2K
POC: Ashish

MISSION LOG

01	Mission Directive: Operation Blind-Nav	2
02	Solar System Reconnaissance Registry	3
03	Technical Contract & Tensor Signatures	5
04	Hardware Qualification Gates	6
05	Live Telemetry & Official Scoring Matrix	7
06	Pre-Flight Transmission Checklist & Conduct	8
G	Participant Guide: Portal, Contract, Submissions & Scoring	9

SECTION 01

01 Mission Directive: Operation Blind-Nav

SOLAR STORM ALERT

✦ INCOMING TELEMETRY: DEEP-SPACE PROBE VANGUARD-IX

STATUS: STAR-TRACKER OFFLINE

ALERT — CORONAL MASS EJECTION IMPACT DETECTED. At 03:14 UTC, a severe solar radiation spike fried Vanguard-IX's primary star-tracker and severed high-gain deep-space uplinks.

The probe is tumbling blind through a gravitational well with orbital decay accelerating. Cloud compute is 400 million kilometers away. Nobody is coming to rescue the craft — you are the **Chief Flight Systems Engineer** at the console, and bringing this probe home is entirely in your hands.

ONLY TWO SYSTEMS REMAIN ONLINE

SYSTEM 01

Nadir optical camera

A downward-facing monochrome camera seeing the surface terrain passing directly beneath the hull.

SYSTEM 02

Flight computer

A radiation-hardened 2GB ARM64 flight computer: a Raspberry Pi 4B.

YOUR TASK

To lock orbital ephemeris, orient the reaction wheels, and fire the return thrusters toward Earth, your onboard neural network must inspect raw **128×128 grayscale** surface terrain passing directly beneath the hull and **classify which celestial body the probe is orbiting** in real time.

THE HARDWARE TRAP

TOO BLOATED

If your model is too bloated, it will trigger an **Out-Of-Memory fault** and brick the flight computer.

TOO COMPRESSED

If you prune or quantize it recklessly to gain speed, crater shadows will blur into noise, accuracy will drop **below 75.0%**, and the probe will fire its thrusters on a fatal trajectory.

Engineer the network, compress the weights, and survive the silicon.

SECTION 02

02 Solar System Reconnaissance Registry

8 CELESTIAL BODIES

TRAINING DATA

All teams train exclusively on the supplied archive:

`astro-geoguessr-train.npz → x_train, y_train`

- Contains **pre-shuffled uint8 nadir patches**.
- Every patch has been mathematically standardized (zero-mean, unit-variance rescaled), so **average pixel brightness is identical across all classes**.
- Your model must learn **true geological morphology** across these 8 solar system bodies.

TARGETS 00 – 03



[00] Moon — LROC

DIAGNOSTIC NADIR GEOMORPHOLOGY

Sharp-rimmed bowl craters, zero atmospheric erosion, mare wrinkle ridges.

COMPRESSION & QUANTIZATION TRAP

Loses crisp ejecta ray contrast under heavy spatial pooling.



[01] Mercury — MDIS

DIAGNOSTIC NADIR GEOMORPHOLOGY

Dense secondary crater chains and long compressional lobate scarps.

COMPRESSION & QUANTIZATION TRAP

95% identical to Lunar terrain unless 1–2px fault scarps are preserved.



[02] Mars — THEMIS

DIAGNOSTIC NADIR GEOMORPHOLOGY

Wind-eroded rims, aeolian dune ripples inside floors, fluidized ejecta.

COMPRESSION & QUANTIZATION TRAP

Aggressive pruning erases fine intracrater sand-dune textures.



[03] Ceres — Dawn FC

DIAGNOSTIC NADIR GEOMORPHOLOGY

Ice-relaxed polygonal/hexagonal crater walls and central sublimation pits.

COMPRESSION & QUANTIZATION TRAP

Angular hexagonal vertices round into circles under low resolution.

SECTION 02

02 Registry, continued

TARGETS 04 – 07



[04] 4 Vesta — Dawn

DIAGNOSTIC NADIR GEOMORPHOLOGY

Equatorial tectonic troughs (Divalia Fossae) and slumped boulder walls.

COMPRESSION & QUANTIZATION TRAP

Naive INT8 quantization bands smooth slope-slump shadow gradients.



[05] 101955 Bennu

DIAGNOSTIC NADIR GEOMORPHOLOGY

Micro-gravity rubble-pile matrix of jagged overlapping dark boulders.

COMPRESSION & QUANTIZATION TRAP

High-frequency boulder edges alias if early conv strides are too wide.



[06] Callisto — Galileo

DIAGNOSTIC NADIR GEOMORPHOLOGY

Saturated ice-rock regolith with bright frost-halo multi-ring fractures.

COMPRESSION & QUANTIZATION TRAP

Concentric shock rings vanish without calibrated activation clipping.



[07] Europa — Galileo

DIAGNOSTIC NADIR GEOMORPHOLOGY

Intersecting tectonic double-ridge ice lineaments and chaos plates.

COMPRESSION & QUANTIZATION TRAP

Criss-crossing 1px ice fractures break apart under unstructured pruning.

CLASS ID QUICK REFERENCE

0 Moon	1 Mercury	2 Mars	3 Ceres
4 4 Vesta	5 101955 Bennu	6 Callisto	7 Europa

SECTION 03

03 Technical Contract & Tensor Signatures

RUNTIME: TFLITE ARM64

✦ ARTIFACT SPECIFICATION: SINGLE-FILE ON-ORBIT DEPLOYMENT

DELEGATE: CPU-ONLY

SINGLE-FILE ARTIFACT

Submit exactly **ONE** self-contained file named `teamname.tflite` (≤ **50.0 MB**).

- All weights, quantization parameters, and graph ops must be embedded.
- Custom Python preprocessing scripts, external lookup tables, or sidecar files are strictly ignored.

TRAINING & RUNTIME ENVIRONMENT

Train remotely on Google Colab, Kaggle, or personal GPUs using TensorFlow/Keras or PyTorch-to-TFLite conversion. Final evaluation runs **strictly offline** on:

Board	Raspberry Pi 4B
CPU	Quad-Core Cortex-A72 @ 1.5 GHz
OS	64-bit
Interpreter	Official CPU interpreter (`ai-edge-litert` / `tflite-runtime`)
Threads	4
Batch size	1

INPUT TENSOR

dtype	**float32**
shape	**[1, 128, 128, 1]**
layout	NHWC: Batch 1, Height 128, Width 128, Channels 1 (grayscale)

The judge harness loads each uint8 test patch, casts to `float32`, and divides by 255.0, so input values lie strictly in **[0.0, 1.0]**.

OUTPUT TENSOR

dtype	**float32**
shape	**[1, 8]**
values	Raw logits or class probabilities for IDs 0 – 7

The judge computes predictions via `argmax(output, axis=1)`.

SECTION 04

04 Hardware Qualification Gates

ZERO-TOLERANCE GATES

✦ ON-BOARD BENCHMARK FILTERS (HIDDEN EASTERN HEMISPHERE TEST SET)

PASS ALL 4 TO RANK

These are survival thresholds. A submission must clear every one of them before it can be ranked.

GATE 01

Navigation Accuracy

≥ 75.0% Top-1 Accuracy

Measured on the hidden evaluation dataset stored in `/opt/dataset` on the Raspberry Pi cluster. Dropping below 75.0% results in immediate mission failure.

GATE 02

Memory Ceiling (RAM)

Peak RSS ≤ 1,024 MiB

Peak Resident Set Size during continuous inference. Linux swap is disabled; exceeding physical memory triggers an instant OOM kill.

GATE 03

Silicon Footprint

Binary ≤ 50.0 MB

Raw `.tflite` binary size, measured prior to loading.

GATE 04

Execution Integrity

Zero failures

No runtime exceptions, missing operator fallbacks, segmentation faults, or tensor shape mismatches across the entire evaluation loop.

SECTION 05

05 Live Telemetry & Official Scoring Matrix

WEIGHTED COMPOSITE

✦ ISOLATED PI-4B BENCH LOOP

1. Cooldown check (< 45°C start)
2. Spawn isolated child process
3. Warm-up pass + timed N=1 loop
4. Push live metrics → leaderboard

Thermal guard: pauses if > 80°C

✦ COMPOSITE SCORE WEIGHTS (100 PTS MAX)

Top-1 Accuracy	**60%**
Inference Speed	**25%**
Peak RSS RAM	**15%**

OFFICIAL FORMULA

$$\text{SCORE} = 100 \times (0.60 \times \text{Norm}_{\text{Acc}} + 0.25 \times \text{Norm}_{\text{Latency}} + 0.15 \times \text{Norm}_{\text{RAM}})$$

NORMALIZATION

Computed among all surviving finalists passing Gates 01 – 04:

Norm_{Acc}	$(\text{Acc}_{\text{team}} - 75.0) / (\text{Acc}_{\text{max}} - 75.0)$
$\text{Norm}_{\text{Latency}}$	$\text{Lat}_{\text{min}} / \text{Lat}_{\text{team}}$
Norm_{RAM}	$\text{RAM}_{\text{min}} / \text{RAM}_{\text{team}}$

MODEL SIZE

Raw model size is strictly enforced via Gate 03 (≤ 50.0 MB), not scored.

TEMPERATURE

SoC temperature is logged continuously to trigger cooldowns (> 80°C) and prevent clock throttling, but is **excluded from direct scoring** to guarantee fairness across boards.

SECTION 06

06 Pre-Flight Transmission Checklist & Conduct

FINAL VERIFICATION

VERIFY LOCAL INTERPRETER EXECUTION

Before uploading to the submission portal:

- [ ] Load your **teamname.tflite** locally using **tf.lite.Interpreter**
- [ ] Allocate tensors
- [ ] Feed a dummy **[1, 128, 128, 1]** **float32** array
- [ ] Verify the output shape is **[1, 8]**
- [ ] Confirm there are no Flex / Select-TF-Ops dependencies

PERMITTED OPTIMIZATION ARSENAL

All of the following are fully encouraged:

- Structured channel pruning
- INT8 Post-Training Quantization (PTQ) with representative calibration data
- Quantization-Aware Training (QAT)
- Knowledge Distillation
- Custom micro-CNN architectures (MobileNet, ShuffleNet, TinyNet)

SUBMISSION LOCK & FAIR PLAY

Only your last valid upload prior to the deadline is ranked on the final leaderboard.

✦ END OF TRANSMISSION ✦

Engineer the network, compress the weights, and survive the silicon.

PART II

EDGE OLYMPICS | PARTICIPANT GUIDE

DATE
30 October 2026

SUBMISSION DEADLINE
6:00 PM
India Standard Time (IST)

OFFICIAL PORTAL
edgeolympics.vercel.app

Build a single TensorFlow Lite image classifier and submit it through your team page. The organizers validate the file, then benchmark eligible models on Raspberry Pi 4B workers against a private evaluation set.

G1 PARTICIPANT GUIDE · SECTION 1 Get started

STEP BY STEP

1. Sign in on the event site. Create a team or join one with the team's invite code. Submit from **My team**.

2. Use the training data and class definitions supplied for the challenge. Keep a separate validation split while developing. **The official evaluation images and labels are private.**

3. Train on your own computer or cloud compute if useful. Only the exported model runs during judging; no notebook, script, network service, or sidecar file accompanies it.

PARTICIPANT GUIDE · SECTION 2

G2 Exact model contract

SPECIFICATION

File	One self-contained `.tflite` file, at most **50 MiB** (50 × 1,048,576 bytes).
Input	One image tensor of shape **[1, 128, 128, 1]**: batch 1, grayscale. Input image values represent the normalized **0–1** range.
Input dtype	**float32, int8, or uint8**. For quantized inputs, the judge uses the model's input scale and zero point.
Output	First output tensor has shape **[1, 8]** and contains eight class scores. The highest score determines the predicted class, indexed **0–7** in the supplied class order.
Runtime	The model must load and run with the official CPU TFLite interpreter on a Raspberry Pi 4B. Avoid custom or delegate-only operators.

Check the exact input/output tensors and run a local inference before uploading. A correct extension alone does not mean the model will pass validation.

PARTICIPANT GUIDE · SECTION 3

G3 Submit and track

PORTAL

HOW TO SUBMIT

Open **My team**, select the `.tflite` file, and press **Submit model** before 6:00 PM IST. The portal displays the deadline in your device's local timezone.

THREE SUBMISSION SLOTS

Each team has **three submission slots**. Pending, running, completed, and disqualified entries use slots. A failed entry releases its slot so you can fix the model and submit again.

RANKING

There is no overwrite or 'latest file wins' behavior. Each upload creates a separate submission; **the best completed eligible score for your team is used for ranking.**

G3 Submit and track, continued

PARTICIPANT GUIDE · SECTION 3

PORTAL

TRACK YOUR RESULT

Follow the result on **My team** or **Leaderboard / My submissions**. Normal progress:

`pending_prelim` → `queued_for_edge` → `running` → `completed`

A failed status includes an error to help diagnose the issue.

G4 How scoring works

PARTICIPANT GUIDE · SECTION 4

LIVE WEIGHTS

ACCURACY

60%

on the hidden set

INFERENCE LATENCY

25%

in milliseconds · ref. 100 ms

PEAK RAM

15%

in MiB · ref. 256 MiB

These are the current weights and reference values. The organizer dashboard can change settings; the live leaderboard shows the active values.

SCORE

$$\text{Score} = 100 \times [\, 0.60 \times \text{accuracy} + 0.25 \times 100 / (100 + \text{latency_ms}) + 0.15 \times 256 / (256 + \text{peak_ram_mb}) \,]$$

Accuracy is a fraction from 0 to 1. Higher score is better. Ties use accuracy (higher), then latency (lower), then RAM (lower).

MODEL FILE SIZE

A submission limit and displayed metric, but not part of the weighted score.

TEMPERATURE

Monitored for safe operation; it is not a scored metric. There is no minimum accuracy qualification threshold.

G5

PARTICIPANT GUIDE · SECTION 5

Before you upload

CHECKLIST

- [ ] Confirm the file is self-contained, under 50 MiB, and has the required shape, dtype, normalized image domain, and class order.
- [ ] Run it locally with a CPU TFLite interpreter and check that all eight output scores are finite.
- [ ] Leave time for validation and, if necessary, another upload before the deadline.

FAIR PLAY

Do not attempt to access the hidden evaluation set, another team's files, or judging infrastructure. Ask an organizer about event logistics or a submission status you cannot resolve.

Portal: edgeolympics.vercel.app | This guide describes the current portal and benchmark contract.







